```
!python --version
```

```
Python 3.12.12
```

Basic Data Types

```
#int
x=3
print(x, type(x))
```

```
3 <class 'int'>
```

```
print(x+1)#addition
print(x-1)
print(x*2)
print(x**2)
print(x)
```

```
4
2
6
9
3
```

```
#float
y= 3.5
print(y, type(y))

print(y, y+1, y*2, y**2)
```

```
3.5 <class 'float'>
3.5 4.5 7.0 12.25
```

```
#boolean
t=True
f=False
# t, f= True, Falase
print(type(t))
```

```
<class 'bool'>
```

```
#String
var1="Hello"
var2="Class"

print(var1, len(var1))
```

```
Hello 5
```

```
new_var= var1 + ' '+ str(x)
print(new_var)
```

```
Hello 3
```

```
another_var='{} {} {}'.format(var2, var1, 12)
print(another_var)
```

```
Class Hello 12
```

```
var3="hello world"
print(var3.capitalize())
print(var3.upper())
print(var3.replace('h','z'))
print(' Data '.strip())
```

```
Hello world
HELLO WORLD
zello world
Data
```

Containers

List, Dictionaries, Set, Tuples

List

```
list1=[3,12,11, 14, 9]
print(list1, list1[2])
```

```
[3, 12, 11, 14, 9] 11
```

```
print(list1[-1])
print(list1[4])#writing at index 2 or fetching data from index 2
print(list1)
```

```
9
9
[3, 12, 11, 14, 9]
```

```
list1[3]='data' # replacing the value present at index 3
print(list1)
```

```
[3, 12, 11, 'data', 9]
```

```
list1.append('Science')
print(list1)
#the size of list is flexible
```

```
[3, 12, 11, 'data', 9, 'Science']
```

```
list1.pop(2)# deletes the last element
print(list1)
```

```
[3, 12]
```

```
numbers=list(range(6))
print(numbers)
```

```
[0, 1, 2, 3, 4, 5]
```

```
#slicing
list2=[11, 2, 5, 18, 3, 20]
print(list2[2:6]) # value at index 2 and 3
print(list2[2:])
print(list2[:4])
print(list2[:])
print(list2[:-2])
list2[2:4]= [50,100]
print(list2)
```

```
[5, 18, 3, 20]
[5, 18, 3, 20]
[11, 2, 5, 18]
[11, 2, 5, 18, 3, 20]
[11, 2, 5, 18]
[11, 2, 50, 100, 3, 20]
```

Dictionary

```
d={'cat':'cute', 'dog':'furry'}#key-value, key can be of any type
print(d['cat'])
print('cat' in d)
```

```
cute
False
```

```
d['dog']='not furry'
print(d)
```

```
{'cat': 'cute', 'dog': 'not furry'}
```

```
del d['cat']
print(d)
```

```
{'dog': 'not furry'}
```

```
Start coding or generate with AI.
```